

Advanced Statistical Mechanics
PHYS6041–Winter, 2026

1. Instructor: Dr. Zong-Chao Yan, Room 229, Department of Physics,
Email: zyan@unb.ca
2. Reference book: “Statistical Mechanics” (2nd ed. 1987, John Wiley & Sons, Inc.) by Kerson Huang
3. **Assessment:**
 - **One mid-term test (40%):** Focuses on core theoretical concepts and their derivations, emphasizing physical insight, mathematical rigor, and the ability to reproduce and clearly explain key results.
 - **Final oral examination (60%):** Assess understanding, synthesis, and the ability to connect core concepts.
4. Office hours: whenever I am free or by appointment via email.
5. Outline:
 - The three laws of thermodynamics
 - Kinetic theory of gases
 - Equilibrium properties of a dilute gas
 - Transport phenomena
 - Classical statistical mechanics
 - Quantum statistical mechanics
6. Honour Statements:

“All work must be your own, and by submitting work in this course, you are indicating that the work is your own, and you are not representing the work of someone else as your own. Getting help on a problem is fine, but the work that you submit must be yours. Copying parts or all of a question from a website or another person is considered plagiarism. If you are unsure if you are committing an academic offense, please ask me about it.”